

CS 2300 – Homework 3

Submit a digital copy (of PDF) to Canvas

Total possible points: 100 points

Homework Guidelines

No Handwritten Answers

This homework will require some written responses and some diagrams. For the written responses, use a standard word processor like Microsoft Word, LibreOffice Writer, Google Docs, or LaTeX. Drawings can be made using Microsoft PowerPoint or Google Drawings¹, among others (e.g. draw.io, lucidchart). You can explore other tools as well, just search around and verify the tool you are considering is compliant with the design aspects that are presented in class.

Please create your answers in a digital format, export to a PDF, and upload that PDF to Canvas. No need to turn a printed copy. For those of you considering LaTeX, ShareLaTeX and Overleaf are good online editors.

One Problem per Page

In the past, many students turned in a single piece of paper with all answered questions smooshed onto it. This did not leave my grader or I with adequate room to provide suggestions on how a design could be improved nor explanations on errors. The process of design work is to work in iterative stages: the first stage will be rather messy (and can be hand-written), the second stage will have things cleaned up, and so on. The final stage is what you should turn in, and it should be free of scratched-out work and high-density drawings. Only answer one problem on *at least* one page, and refrain from super-compact answers that leaves more than 1/2 of the page empty. Ideally, leave somewhere between 1/4 to 1/2 of a page empty so that we can make annotations.

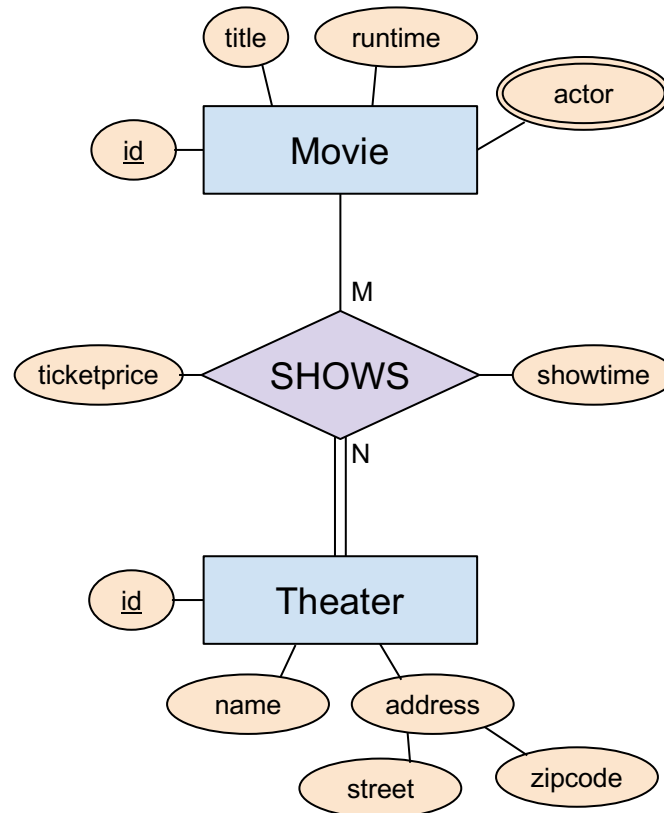
¹ <https://docs.google.com/drawings/>

Problem 1 [10 pts]

Convert the following ER diagram into relational schemas.

Make sure to denote all primary keys and foreign keys by drawing the arrows and include the data types for each attribute, e.g. R(id:int, name:str, ...).

Artificial keys should only be used when absolutely necessary. If a natural key exists and an artificial key is used instead, **points will be deducted**.



Problem 2 [40 pts]

Convert the following EER diagram into a relational database schema.

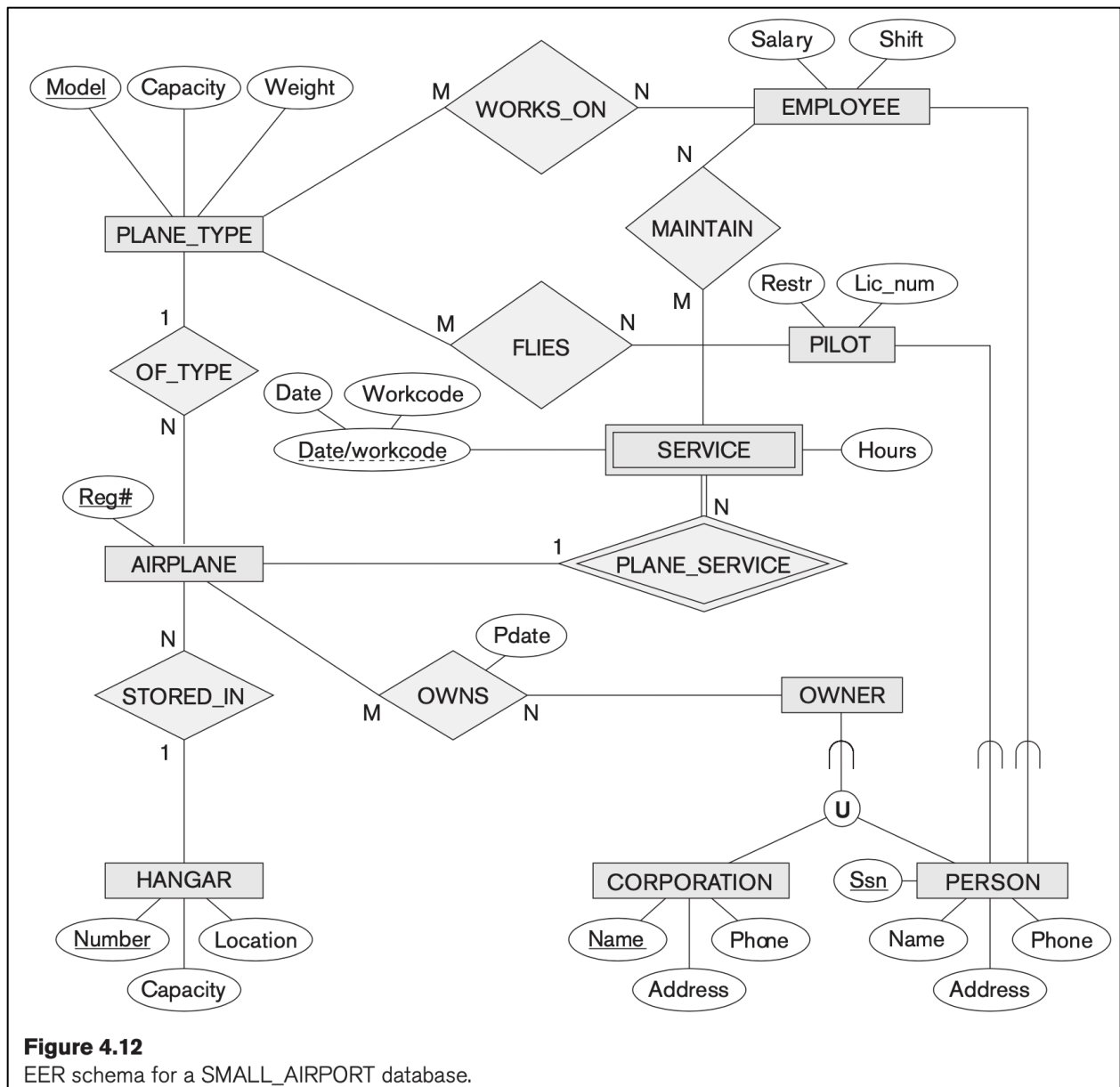
Figure 4.12 shows an example of an EER diagram for a small-private-airport database; the database is used to keep track of airplanes, their owners, airport employees, and pilots.

From the requirements for this database, the following information was collected:

- Each AIRPLANE has a registration number, is of a particular plane type, and is stored in a particular hangar.
- Each PLANE_TYPE has a model number, a capacity, and a weight.
- Each HANGAR has a number, a capacity, and a location.
- The database also keeps track of the OWNERS of each plane and the EMPLOYEES who have maintained the plane.
- Each relationship instance in OWNS relates an AIRPLANE to an OWNER and includes the purchase date.
- Each relationship instance in MAINTAIN relates an EMPLOYEE to a service record.
- Each plane undergoes service many times.
- A SERVICE record includes as attributes the date of maintenance, the number of hours spent on the work, and the type of work done.
- Airplane service is represented as a weak entity type because the airplane registration number is used to identify a service record.
- An OWNER is either a person or a corporation, thus represented using a union type.
- Both pilots and employees are subclasses of PERSON.
- Each PILOT has specific attributes license number and restrictions.
- Each EMPLOYEE has specific attributes salary and shift worked.
- All PERSON entities in the database have data kept on their Social Security number, name, address, and telephone number.
- For CORPORATION entities, the data kept includes name, address, and telephone number.
- The database also keeps track of the types of planes each pilot is authorized to fly and the types of planes each employee can do maintenance work on.

Make sure to denote all primary keys and foreign keys by drawing the arrows and include the data types for each attribute, e.g. R(id:int, name:str, ...).

Artificial keys should only be used when absolutely necessary. If a natural key exists and an artificial key is used instead, **points will be deducted**.



Problem 3 [50 pts]

Convert the following EER diagram into a relational database schema.

The figure below shows an example of an EER diagram for the Japanese Vanisher Company. It is a secretive and highly specialized organization that helps individuals disappear from society for various reasons, whether they are escaping danger, starting a new life, or hiding from threats. The company provides clients with safe houses, new identities, and necessary survival tools while ensuring their tracks are erased. This database is designed to manage the logistics of these operations.

From the requirements for this database, the following information was collected:

- A **Client** is assigned a **Vanisher** to help them disappear.
- Both **SafeHouse** and **DisappearanceKit** belong to the **Resource** category/union type.
- A Client undergoes **Operations**, involving exactly ONE Client and at least ONE or MORE vanishers. A safe house is optional—some clients may disappear without using one.
 - Each operation must have a specific **Date**.
 - A client CANNOT have multiple operations on the same day.

Make sure to denote all primary keys and foreign keys by drawing the arrows and include the data types for each attribute, e.g. R(id:int, name:str, ...).

Artificial keys should only be used when absolutely necessary. If a natural key exists and an artificial key is used instead, **points will be deducted**.

